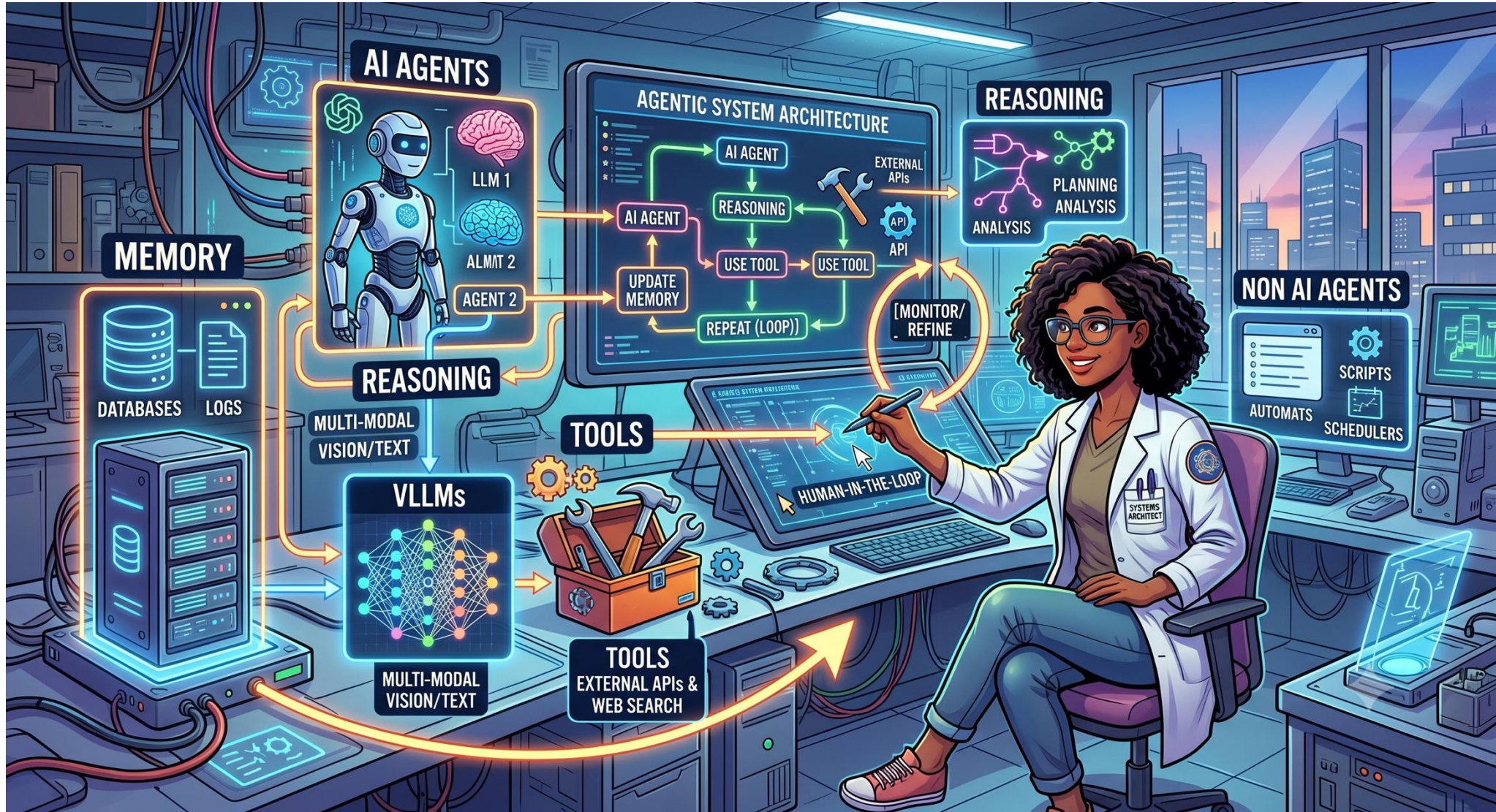


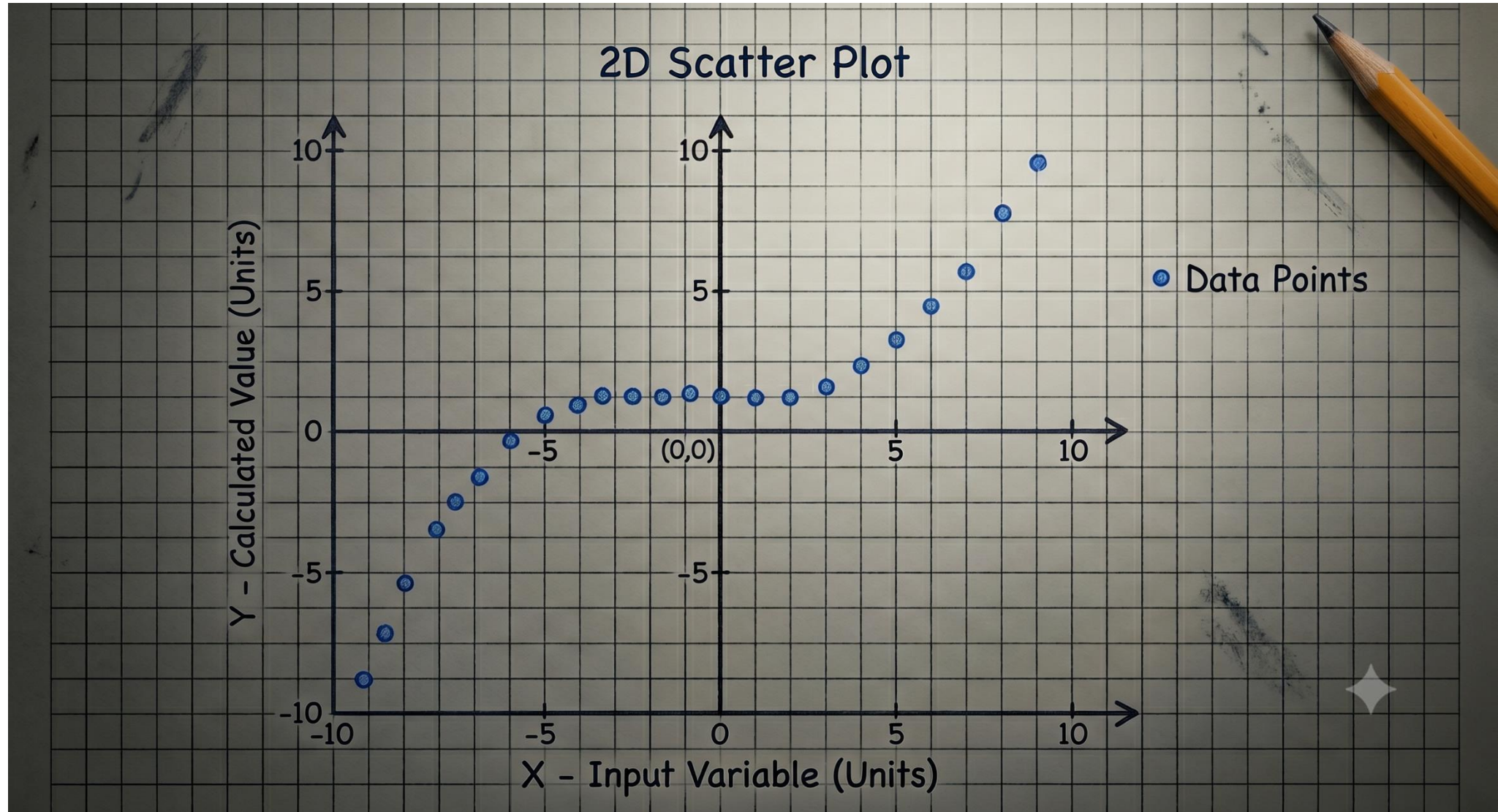
Building your own agentic system



Why building your own agentic system?

- You control the flow, tools, and depth
- You control the LLMs to use (open or close source)
- You need to run the system many times and need a reproducible pipeline
- Control what is in the context of each agent
- We have control over applications to science
- Is fun!

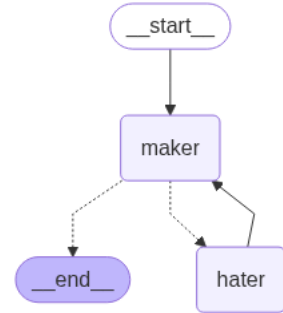
Task

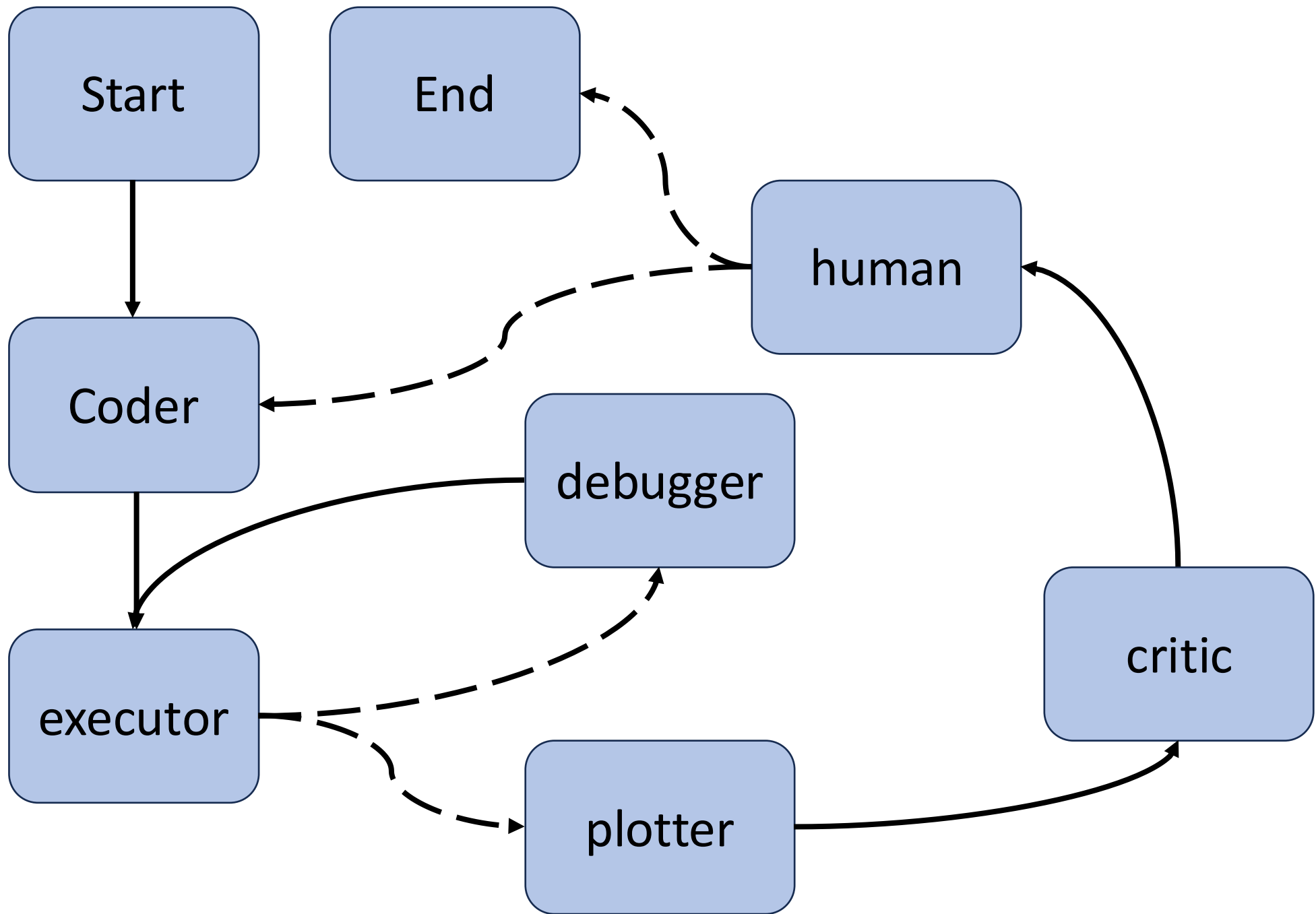


Ingredients

- AI agents / Non-AI agents
- Tools
- Memory
- VLLMs
- Loops
- Human-in-the-loop

LangGraph

<u>Agents</u>	<u>Graph State</u>	<u>Computational Graph</u>
• <i>Task</i> • <i>Input</i> • <i>Output</i>	<i>Centralized register for messages</i> <i>Used to manage communication among agents</i> <i>Set agent context and memory</i>	<i>Set system workflow.</i> <i>Can be static or dynamic</i>  <pre>graph TD; start([__start__]) --> maker[maker]; maker -.-> end([__end__]); maker --> hater[hater]; hater --> maker</pre>



https://github.com/franciscovillaescusa/AI-Science_Chicago_2026

<https://tinyurl.com/5c2j7dv4>

Discussion / Poll time